CampusX ML Beginner Notes (Original) Lectures 1–5

Lecture 1: What is Machine Learning?

Goal: Understand the core concept.

Explanation: ML enables computers to learn patterns from data and make predictions without being explicitly programmed.

Key Points: AI, ML, DL overview

Real-world Examples: Spam filtering, recommendations, fraud detection.

Python Libraries: NumPy, Pandas, Scikit-learn (where applicable).

Practice: Revise the concept and try one small coding example.

Lecture 2: AI vs ML vs Deep Learning

Goal: Understand the core concept.

Explanation: AI is the broad field, ML is learning from data, DL uses deep neural networks.

Key Points: Relationship between AI, ML and DL

Real-world Examples: Image recognition, chatbots.

Python Libraries: NumPy, Pandas, Scikit-learn (where applicable).

Practice: Revise the concept and try one small coding example.

Lecture 3: Types of Machine Learning

Goal: Understand the core concept.

Explanation: Supervised, Unsupervised, Semi-supervised and Reinforcement Learning.

Key Points: Labels, clustering, rewards

Real-world Examples: Classification and regression.

Python Libraries: NumPy, Pandas, Scikit-learn (where applicable).

Practice: Revise the concept and try one small coding example.

Lecture 4: Batch Learning

Goal: Understand the core concept.

Explanation: Models are trained on the full dataset and updated periodically.

Key Points: Offline training

Real-world Examples: Suitable for static datasets.

Python Libraries: NumPy, Pandas, Scikit-learn (where applicable).

Practice: Revise the concept and try one small coding example.

Lecture 5: Online Learning

Goal: Understand the core concept.

Explanation: Model learns continuously from incoming data.

Key Points: Incremental learning

Real-world Examples: Useful for streaming data.

Python Libraries: NumPy, Pandas, Scikit-learn (where applicable).

Practice: Revise the concept and try one small coding example.